

Answer key Exercise HCC-1: Temperature Conversions

Source: Solution and Answer Guide, Silberstein/Obrzut, The Complete HVACR Manual 2E, Section 1: HVACR Core Concepts (HCC), and the HCC-1 lab sheet.

Context

Objectives: Upon completion of this exercise, you should be able to convert temperatures from one scale to another using mathematical equations.

Introduction: Using a calculator and the appropriate mathematical relationships, you will complete the temperature charts provided.

Takeaway: There are many different scales used to measure temperature and their uses vary from one application to the next. A technician must be proficient at using these scales, understanding absolutes, and converting from one scale to another.

Exercise HCC-1

HCC-1 Step-by-step procedure, temperature chart

Using your calculator or conversion app, complete the following chart by determining the correct values for all of the missing temperature readings.

Model answer: Completed temperature conversion chart. The column "Given" shows the value that was printed on the chart; the student fills in the other three.

Row	Given	Fahrenheit	Celsius	Rankine	Kelvin
1	Rankine 0	-460	-273	0	0
2	Fahrenheit 32	32	0	492	273
3	Celsius 100	212	100	672	373
4	Rankine 600	140	60	600	333
5	Kelvin 0	-460	-273	0	0
6	Fahrenheit 200	200	93	660	366
7	Rankine 100	-360	-218	100	55
8	Celsius 50	122	50	582	323
9	Kelvin 300	81	27	541	300
10	Celsius 0	32	0	492	273
11	Fahrenheit 150	150	66	610	339
12	Rankine 50	-410	-246	50	27
13	Celsius -40	-40	-40	420	233
14	Fahrenheit -40	-40	-40	420	233
15	Fahrenheit 212	212	100	672	373
16	Kelvin 373	212	100	672	373
17	Celsius -273	-460	-273	0	0
18	Rankine 672	212	100	672	373
19	Fahrenheit 5	5	-15	465	258
20	Kelvin 500	441	227	901	500

HCC-1 Summary statement

Explain the differences between an absolute temperature scale, such as the Rankine scale, and a non-absolute temperature scale, such as the Fahrenheit scale. In your response, be sure to include the significance of the low-end values of the scales.

Model answer: Kelvin and Rankine are absolute scales; Fahrenheit and Celsius are not. On an absolute scale, zero is absolute zero, so the scale has no negative values: 0°R and 0 K are the same point as -460°F and -273°C . A non-absolute scale puts its zero at another point (0°C is 32°F , 492°R and 273 K) and uses negative values below it, down to -460°F and -273°C at absolute zero.

Not in the official answer key. Composed only from model answer 2 and the completed temperature chart (rows 1, 5, 10 and 17).

Question HCC-1.1

The mathematical relationship between a Fahrenheit temperature and the equivalent Celsius temperature is:

- A. 1:1.
- B. 2:1.
- C. 3:1.
- D. It cannot be determined, as the values are not proportional.

Model answer: D. It cannot be determined, as the values are not proportional.

Question HCC-1.2

Which of the following temperature scales represent absolute values?

- A. Celsius and Kelvin
- B. Fahrenheit and Celsius
- C. Kelvin and Rankine
- D. Fahrenheit and Kelvin

Model answer: C. Kelvin and Rankine

Question HCC-1.3

Which of the following temperatures represents absolute zero?

- A. -460°F
- B. 0°C
- C. 672°R
- D. 273°K

Model answer: A. -460°F

Not in the official answer key, which stops after question 2. Derived from the completed temperature chart, row 1: $-460^{\circ}\text{F} = 0^{\circ}\text{R} = 0\text{ K}$.

Question HCC-1.4

Which of the following temperatures is the coldest?

- A. 32°F
- B. 672°R
- C. 100°C
- D. 373°K

Model answer: A. 32°F

Not in the official answer key. Derived from the completed temperature chart: $32^{\circ}\text{F} = 0^{\circ}\text{C}$ (row 2), while 672°R , 100°C and 373 K are all 212°F (rows 3 and 18).

Question HCC-1.5

Which of the following temperatures is the hottest?

- A. 212°F
- B. 492°R
- C. 273°K
- D. 0°C

Model answer: A. 212°F

Not in the official answer key. Derived from the completed temperature chart: $212^{\circ}\text{F} = 100^{\circ}\text{C}$ (row 15), while 492°R , 273 K and 0°C are all 32°F (rows 2 and 10).

Question HCC-1.6

Which of the following temperatures represent the same level of heat intensity?

- A. 0°C and 0°K
- B. 0°F and 0°R
- C. -40°C and -40°F
- D. 50°C and 50°R

Model answer: C. -40°C and -40°F

Not in the official answer key. Derived from the completed temperature chart, rows 13 and 14: $-40^{\circ}\text{C} = -40^{\circ}\text{F}$.